DocInsights Workshop at EMNLP 2026
Shared Task, Dr.DocBench Track

# Consensus Selection over Prompt-Diverse VLM Transcriptions for Expert-Level Document Parsing

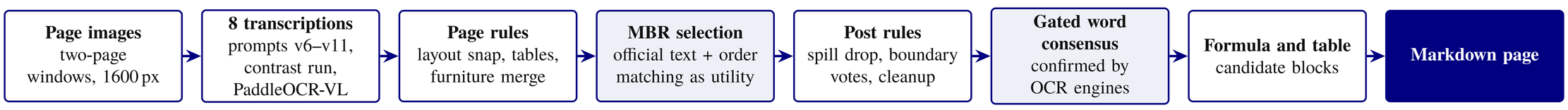
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Scan for the full report
pranshurastogi29.github.io/drdocbench

| 7th | 84.54 | 0.1641 | 85.65 | 81.84 | 71.86 |
|---|---|---|---|---|---|
| final leaderboard | Overall | Text_ED | Reading order | Formula CDM | Table TEDS |

509 test pages · training-free · one free vision-language model, prompted many ways · about $82 of API use in total




## 1 The Challenge

Dr.DocBench (Yang et al., 2026), a track of the DocInsights 2026 shared task (DocInsights Workshop Organizers, 2026), collects book pages on which parsers disagree. Its references follow **book-specific conventions** (which lines form a block, whether picture text is kept, where page numbers go), so a strong model loses points on pages it reads correctly.
Scoring uses the md2md matcher of OmniDocBench (Ouyang et al., 2025): text edit distance, reading order (RO), CDM (Wang et al., 2025) and TEDS (Zhong et al., 2020); Overall averages them per page.

**Scorer facts that shaped the system**

- Predicted blocks are joined to one reference block only if it has at most 300 normalised characters.
- Only word characters count: punctuation and Markdown emphasis do not; case and digits do.
- Formulas and tables are scored by best match, apart from text and order.

## 2 Select, Don't Train

One model gives many plausible transcriptions. We pick one per page with minimum Bayes risk (Kumar and Byrne, 2004), using the **metric itself as the utility**:

$$\hat{c} = \arg\max_{c \in C} \frac{1}{|R_c|} \sum_{r \in R_c} \left( w_t \, T(c \mid r) + w_o \, O(c \mid r) \right)$$

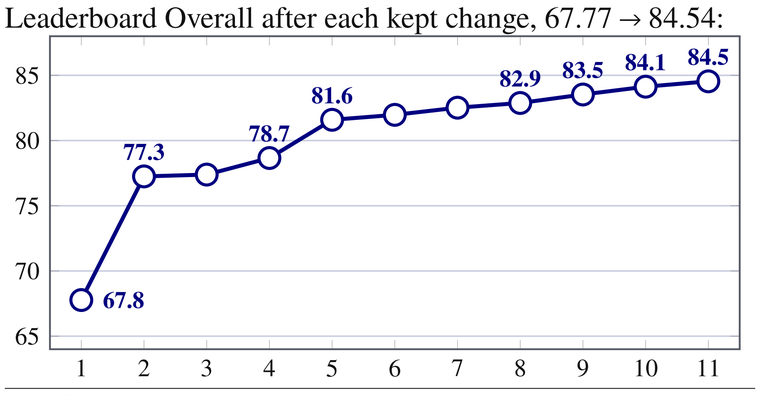
$T$ and $O$ are the official text and order scores of $c$ with run $r$ as the reference, $R_c$ excludes $c$, and $(w_t, w_o) = (0.558, 0.433)$, the weights of text and order in Overall fitted on our submissions.
**mbr8**: eight candidates (five base runs incl. PaddleOCR-VL (Cui et al., 2025), three column-break runs) against eleven pseudo-references.

## 3 Gated Word Consensus

In the spirit of ROVER (Fiscus, 1997), the other runs vote word by word. A word changes only if a strict majority agrees, the new word is a near-spelling of the old one (not case only, not at a paragraph edge), and PaddleOCR-VL or MinerU (Wang et al., 2024) confirms it or the old word is not in a dictionary.
A plain majority vote lost **0.06**: our runs share one model's errors. The gated vote gains **+0.006** with no page worse and fixes 33 test words, e.g. *ATHAWAY* → *HATHAWAY*.

## 4 Development Journey

Leaderboard Overall after each kept change, 67.77 → 84.54:




| # | Change kept | Overall |
|---|---|---|
| 1 | Single run, prompt v3 | 67.77 |
| 2 | Furniture-last prompts, medoid of runs, layout snap | 77.25 |
| 3–4 | PaddleOCR-VL tables; contrast run as a vote | 78.66 |
| 5–7 | Formula and table candidate blocks | 82.52 |
| 8 | **MBR selection**, furniture merge, spill drop, boundary votes | 82.88 |
| 9 | More formula candidates, symbol channel | 83.53 |
| 10 | **Column-break runs in MBR**, word consensus | 84.13 |
| 11 | **Final**: more candidate sources | **84.54** |

## 5 Ablations

Leave-one-out on 925 development pages (Δ in leaderboard units):

| Pipeline | Text_ED | RO | Δ |
|---|---|---|---|
| Final (mbr8 + rules) | 0.1168 | 88.58 | – |
| – layout snap, tables | 0.1204 | 87.11 | **−0.84** |
| – furniture merge | 0.1162 | 87.49 | **−0.44** |
| – boundary votes | 0.1186 | 88.48 | **−0.14** |
| – spill drop | 0.1187 | 88.55 | **−0.12** |
| – cleanup | 0.1169 | 88.58 | **−0.01** |
| + word consensus | 0.1168 | 88.58 | **+0.005** |

**Selection**

- MBR beats the medoid by **+1.05** (book-clustered CI [+0.38, +1.85]).
- Learned rankers, weighted MBR and paragraph fusion: within noise of MBR.
- A per-page oracle over the same runs adds +4.8: selection, not reading, is the limit.
- Column-break runs as candidates: **+0.88** on dev, **+0.60** on the leaderboard.

## 6 What Transferred to Test

**Selection did**: mbr8 kept 68% of its development gain (94% for reading order). **Convention rules did not**:

| Change (Overall) | Dev | Test |
|---|---|---|
| Contrast runs as extra votes | **+0.39** | **−0.84** |
| One-page windows as the pool | **+1.02** | **−0.63** |
| GPT-6.1 on 120 hard pages | **+0.06** | **−0.47** |
| Drop text PaddleOCR-VL lacks | **+0.02** | **−0.21** |
| Column splits incl. lists | **+1.00** | **−0.02** |

## 7 Where the Error Is

Diagnostic submissions blanked groups of test books; the score change decodes each group:

| Test books | Pages | Text_ED | RO |
|---|---|---|---|
| Subjects absent from dev | 120 | 0.200 | 77.4 |
| Highest run disagreement | 120 | 0.203 | 80.7 |
| Half of the rest | 134 | 0.157 | 85.0 |
| Other half | 135 | 0.120 | 87.5 |

The first two groups hold **56%** of the text error. On dev, half of it sits on 8% of pages with a book-specific reference choice.

## 8 Stronger Models Alone

| 53 common dev pages | Text_ED | RO | Δ |
|---|---|---|---|
| Our pipeline (mbr8) | 0.0857 | 88.45 | – |
| Kimi K2.5 | 0.0614 | 85.20 | **−0.05** |
| Gemini 3.8 Flash | 0.0682 | 84.90 | **−0.56** |
| Kimi K2.6, no thinking | 0.1053 | 83.71 | **−3.14** |
| GPT-6 sol-pro (897 pages) | 0.1380 | 83.69 | **−3.19** |

Kimi K2.5 (Kimi Team et al., 2026) reads text best but orders worse; proofreading our pages with it: **−2.6**.

**Takeaways**

- Prompt diversity plus MBR with the metric as utility beat every stronger single model we tried.
- Selection transfers to the hidden test set; rules tuned to development books do not.
- The remaining gap is in segmentation and order conventions that vary by book.

**Models and tools.** Base transcriber: a free OpenRouter model (OpenRouter, 2026b). Also in the final submission: GPT-6.1 sol-pro (OpenAI, 2026c), as a source of formula and table candidates. Tested: GPT-6 sol-pro and luna-pro (OpenAI, 2026b,a), Gemini 3.8 Flash (Google, 2026), Claude Sonnet 5.5 (Anthropic, 2026), Kimi K2.6 (Moonshot AI, 2026), Qwen3.8 (Qwen Team, 2026), GLM-5.3 Flash (Z.ai, 2026), DeepSeek V4.1 Flash (DeepSeek, 2026), MiMo v2.6 Pro (Xiaomi, 2026) and dots-3-note-preview (dots studio, 2026), through OpenRouter (OpenRouter, 2026a) and Hugging Face Inference Providers (Hugging Face, 2026). Training-free: no fine-tuning, no external annotated data.

**Links.** Final ranking: https://drdocbench-challenge.abaka-pages.com/#leaderboard · Report: https://pranshurastogi29.github.io/drdocbench/ · Paper: https://doi.org/10.13140/RG.2.2.19858.21444 · Code (MIT): https://github.com/pranshurastogi29/drdocbench-challenge

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